CNN-Powered Waste Classification

Smart Trash Detection AI

Upload any image and our CNN model instantly identifies the waste type and tells you exactly how to recycle it.

🔍 Detect Trash → How It Works

6	74%+	CNN	5
Waste Categories	Model Accuracy	MobileNetV2	Recyclable Types

CATEGORIES

6 Waste Categories

Our model classifies waste into 6 categories with recycling guidance for each.

Cardboard

Boxes, packaging, paper tubes. Flatten before recycling.

♻ Recyclable

Glass

Bottles and jars. Rinse and separate by color.

♻ Recyclable

Metal

Aluminum cans, tins. Highly valuable recyclable material.

♻ Recyclable

Paper

Newspapers, magazines, office paper. Keep dry.

♻ Recyclable

Plastic

Bottles, containers. Check number (1-7) for recycling rules.

♻ Recyclable

Trash

General waste. Cannot be recycled — goes to landfill.

✕ Not Recyclable

PROCESS

How It Works

01

Upload Image

Take a photo or upload an existing image of any waste item from your device.

02

CNN Analysis

MobileNetV2 CNN model analyzes the image and classifies the waste type instantly.

03

Get Guidance

Receive recycling instructions, disposal tips, and environmental impact info.

TECH STACK

Built With

- Python
- TensorFlow / Keras
- MobileNetV2 CNN
- Flask
- SQLite
- HTML5 / CSS3
- JavaScript
- TrashNet Dataset
- Google Colab

Classify Your Waste Now

Upload any image and get instant recycling guidance powered by deep learning.

Start Detection →